

Team Name : Antariksh Architects

Team Leader Name : Kalpesh Sunil Patil

Problem Statement :Chandrayaan-2: Subsurface Ice Detection & Lunar Landing Planning[PS-8]

Team Members

Team Leader: Name: Kalpesh Sunil Patil College: S.V.K.M IOT Dhule	Team Member-1: Name: Tarun Hemraj Chaudhari College: S.V.K.M IOT Dhule
Team Member-2: Name: Dhanashri Bhatu Patil College: S.V.K.M IOT Dhule	Team Member-3: Name: Gaurav Manoj Chaudhari College: S.V.K.M IOT Dhule

Opportunity & Innovation

Existing Challenges

- ❌ Uncertain subsurface ice locations in PSRs
- ❌ Difficult landing decisions near shadowed craters
- ❌ Hazardous rover traversal through rough terrain
- ❌ Rover traversal paths planned manually - high risk
- ❌ Limited estimation of extractable ice resources

Our AI-Powered Innovation

- ✓ ML-based subsurface ice probability mapping from DFSAR data
- ✓ Automated hazard detection using OHRC and DTM fusion
- ✓ A* Algorithm for optimal, safe rover navigation paths
- ✓ Intelligent landing site ranking with multi-criteria scoring

USP:ML-based mission planning using Chandrayaan-2 DFSAR data for landing, driving, excavation, and ice estimation.

Proposed Solution Features

Seven integrated capabilities spanning the full mission lifecycle — from orbital data ingestion to autonomous surface operations.

AI Ice Detection

CatBoost model on DFSAR polarimetric data for subsurface ice probability

Terrain Analysis

Slope, roughness, and elevation maps from OHRC and DTM fusion

Hazard Detection

Automated crater, boulder, and slope hazard zone identification

Landing Recommendation

Multi-criteria ranked safe zones with confidence scoring

A* Rover Navigation

Optimal traversal paths avoiding hazards, minimizing energy cost

Ice Volume Estimation

Quantified subsurface ice deposits for in-situ resource utilization planning

Mission Intelligence Dashboard

Unified interface for ice maps, hazard layers, and rover

Process Flow Diagram

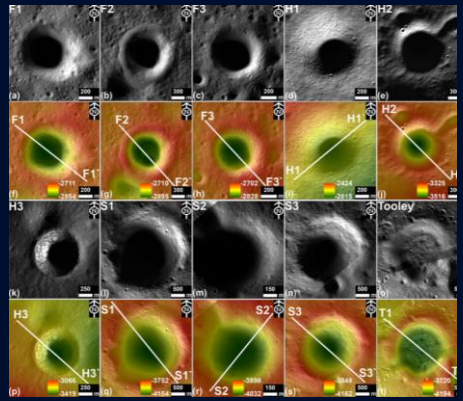
Mission Objectives

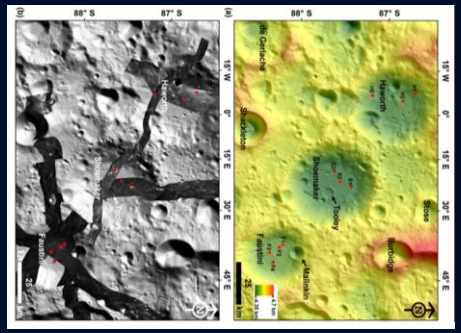
🎯 **Where to Land**
Safe landing zone identification using terrain hazard analysis.

🚗 **Where to Drive**
Optimal rover traverse planning using A* algorithm.

⛏️ **Where to Excavate**
Selection of high ice probability excavation sites.

🧊 **How Much Ice is Available**
Subsurface ice volume estimation and characterization.





DFSAR + DTM + Orthophoto Data

Data Verification

Feature Extraction (HH,HV,VH,VV,Incident Angle

CatBoost Ice Classification

Ice Detection Heatmap

Safe Landing Zone Selection

A* Rover Traverse Planning

Excavation Target Selection

Ice Volume Estimation

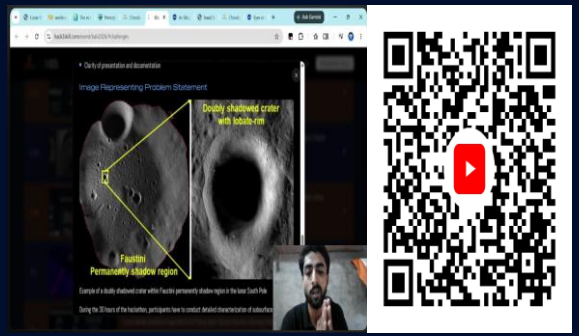
Mission Dashboard

Wireframes/Mock diagrams of the proposed solution

Dashboard



Scan to view our project demo



Ice Detection



Hazard Analysis



Landing Sites



Terrain Analysis



Rover Navigation

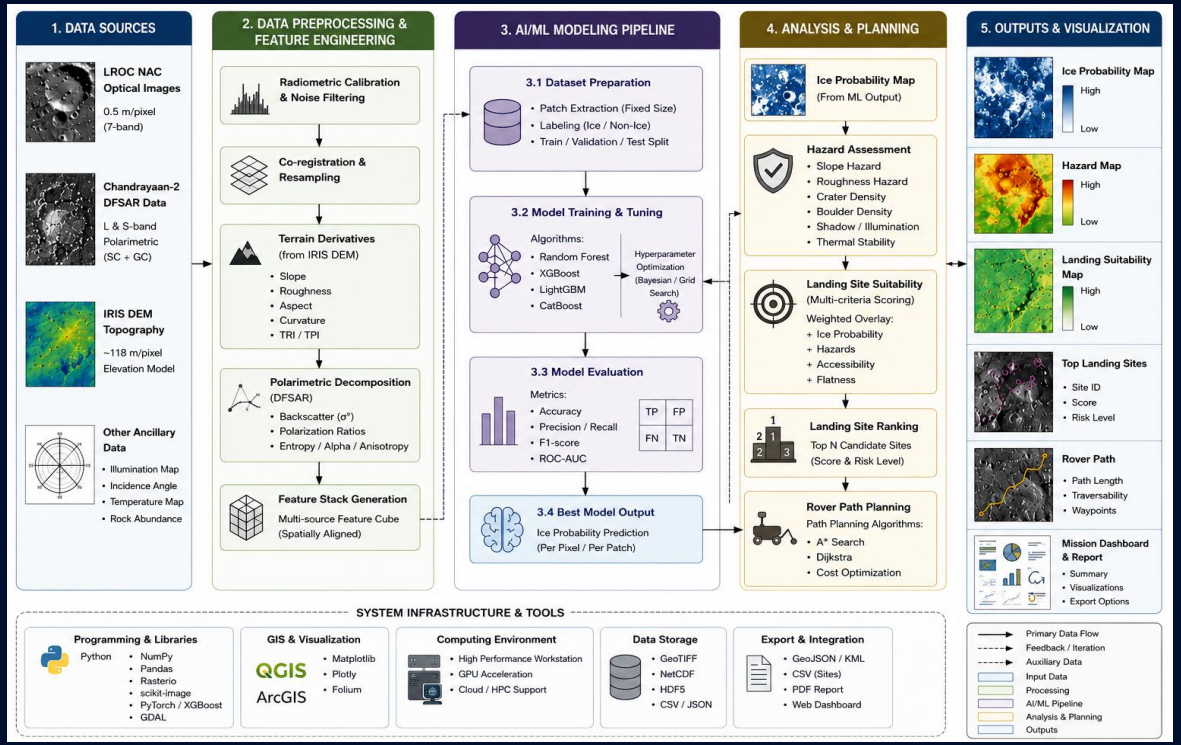


Architecture diagram of the proposed solution

A modular AI-driven architecture that integrates Chandrayaan-2 datasets to perform terrain analysis, ice detection, hazard mapping, landing site recommendation, and A rover path planning for intelligent lunar mission support.*

Key Features

- Modular and scalable architecture.
- Multi-source data integration (DFSAR, DTM, OHRC).
- AI-powered decision support.
- Automated terrain and hazard analysis.
- A* based autonomous rover navigation.
- Interactive mission visualization.



Technologies Used

Full stack — languages, frameworks, datasets, cloud & tools

Machine Learning

- Python
- CatBoost
- XGBoost
- Scikit-learn

Geospatial Processing

- Rasterio
- OpenCV
- GeoPandas
- GDAL

Scientific Computing

- NumPy
- SciPy
- Matplotlib
- Pandas

Frontend

- React Js
- Next Js
- Tailwind CSS
- TypeScript

Mission Planning

- A* Path Analysis
- Terrain Analysis
- Landing Site Optimization

Visualization

- QGIS
- Leaflet
- Matplotlib

Deployment & Cloud Infrastructure

- AWS EC2
- AWS S3
- AWS CloudFront
- Docker
- Git & GitHub

Backend

- Fastapi
- Python
- Rest APIs

Estimated implementation cost

Estimated cost scaling the proposed lunar mission planning system to operational deployment.

Cloud Infrastructure

- AWS EC2, S3, RDS
- Storage & Backup
- Networking & Security

₹10,00,000

AI Compute Resources

- GPU Instances
- Model Training & Inference
- High Performance Computing

₹15,00,000

Development Team

- 5 Engineers
- 18 Months of Development
- DevOps & Project Management

₹30,00,000

Geospatial Data Processing

- Data Acquisition & Licensing
- Geospatial Processing
- Storage & Management

₹10,00,000

Testing & Validation

- System Testing
- Model Validation
- Performance Evaluation

₹10,00,000

Deployment & Maintenance

- Deployment
- Monitoring & Maintenance
- Updates & Support

₹7,00,000

Total Estimated Budget ₹80,00,000

Resources: 5 ML Engineers • 1 DevOps • 1 Geospatial Analyst • 8× NVIDIA A100 GPUs • AWS Tenancy • ISRO MOSDAC API Access


BHARATIYA
ANTARIKSH
HACKATHON
2026
THANK YOU